The Collected Mathematical Papers of Arthur Cayley

Vol. XII (Scan PDF pp. 226–237; book pp. 206–217)

Paper 825 (Memoir on the Abelian and Theta Functions, concl. of Ch. VI; Ch. VII—The Functions T, U, V, Θ); Paper 826 (Note on a Partition-Series).

If, reverting to the identity, we write therein for instance $x = a$, we find

$$\alpha a^3 + \beta a^2 + \gamma a + \delta = \sqrt{\mu}\, A_a A_m A_m,$$

which equation when properly reduced gives the proportional value of A_m.

162. Calling for a moment the function on the left-hand side Ω, we have

$$\Omega \;=\; \begin{vmatrix} x_1^3 & x_1^2 & x_1 & 1 & \sqrt{\lambda}\sqrt{X_1} \\ x_2^3 & x_2^2 & x_2 & 1 & \sqrt{\lambda}\sqrt{X_2} \\ x_3^3 & x_3^2 & x_3 & 1 & \sqrt{\lambda}\sqrt{X_3} \\ x_4^3 & x_4^2 & x_4 & 1 & \sqrt{\lambda}\sqrt{X_4} \\ a^3 & a^2 & a & 1 & \Omega \end{vmatrix} \;=\; 0,$$

that is,

$$\Omega \begin{vmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \\ x_4^2 & x_4 & 1 \end{vmatrix} \;+\; \sqrt{\lambda} \begin{vmatrix} x_1^3 & x_1^2 & x_1 & \sqrt{X_1} \\ x_2^3 & x_2^2 & x_2 & \sqrt{X_2} \\ x_3^3 & x_3^2 & x_3 & \sqrt{X_3} \\ x_4^3 & x_4^2 & x_4 & \sqrt{X_4} \end{vmatrix} = 0,$$

viz. this is

$$\begin{aligned} &\Omega\,(x_1 - x_2)(x_1 - x_3)(x_1 - x_4)(x_2 - x_3)(x_2 - x_4)(x_3 - x_4) \\ = \; -\sqrt{\lambda}\,\{ &\sqrt{X_1} \cdot x_2 - x_3 \cdot x_2 - x_4 \cdot x_3 - x_4 \cdot a - x_1 \cdot x_2 - a \cdot x_3 - a \cdot x_4 - a \\ + &\sqrt{X_2} \cdot x_1 - x_3 \cdot x_1 - x_4 \cdot x_3 - x_4 \cdot a - x_2 \cdot x_1 - a \cdot x_3 - a \cdot x_4 - a \\ + &\sqrt{X_3} \cdot x_1 - x_2 \cdot x_1 - x_4 \cdot x_2 - x_4 \cdot a - x_3 \cdot x_1 - a \cdot x_2 - a \cdot x_4 - a \\ + &\sqrt{X_4} \cdot x_1 - x_2 \cdot x_1 - x_3 \cdot x_2 - x_3 \cdot a - x_4 \cdot x_1 - a \cdot x_2 - a \cdot x_3 - a \};\end{aligned}$$

or, as this may be written,

$$\begin{aligned} &\Omega \,.\, x_1 - x_2 \,.\, x_1 - x_3 \,.\, x_2 - x_3 \,.\, x_1 - x_4 \,.\, x_2 - x_4 \,.\, x_3 - x_4 \\ = \; &\frac{\sqrt{\lambda} \,.\, a - x_1 \,.\, a - x_2}{x_1 - x_2} \left[x_3 - x_2 \,.\, x_3 - x_4 \,.\, a - x_3 \,.\, \sqrt{X_4} - (x_4 - x_2 \,.\, x_4 - x_3 \,.\, a - x_4 \,.\, \sqrt{X_3})\right] \\ + &\frac{\sqrt{\lambda} \,.\, a - x_1 \,.\, a - x_4}{x_2 - x_4} \left[x_2 - x_1 \,.\, x_2 - x_3 \,.\, a - x_2 \,.\, \sqrt{X_3} - (x_3 - x_1 \,.\, x_3 - x_2 \,.\, a - x_3 \,.\, \sqrt{X_2})\right].\end{aligned}$$

We have here $\sqrt{\lambda} \,.\, a - x_1 \,.\, a - x_2 = \sqrt{\lambda}\, A_a^2 A_m^2$, and the function

$$\frac{1}{x_3 - x_4}\left[x_2 - x_3 \,.\, x_2 - x_4 \,.\, b_0 \sqrt{X_4} - (x_1 - x_3 \,.\, x_1 - x_4 \,.\, b_0 \sqrt{X_3})\right],$$

which multiplies this, is without difficulty found to be

$$= \frac{-A_{12}}{c - d \,.\, d - b \,.\, b - c}\, \Sigma\, [c - d \,.\, B_{12}^x B E_a \,.\, C_a \,.\, D_a],$$

where the summation extends to the three terms obtained by the cyclical interchange of the letters b, c, d: these being a set of three out of the five letters other than a. Similarly $\sqrt{\lambda}\,.\,a-x_1\,.\,a-x_2=\sqrt{\lambda}\,A_a^2A_m^2$, and the function which multiplies this is

$$= \frac{-A_{12}}{c-d\,.\,d-b\,.\,b-c}\,\Sigma\,[c-d\,.\,B_{12}^x\,.\,BE_aC_aD_a].$$

The expression for Ω thus contains the factor $A_{12}A_{34}$: but we have

$$\Omega=\alpha a^3+\beta a^2+\gamma a+\delta=\sqrt{\mu}\,A_aA_mA_m;$$

this equation contains therefore the factor $A_{12}A_{34}$, and omitting it we find

$$-\frac{\sqrt{\mu}}{\sqrt{\lambda}}\,(x_1-x_2\,.\,x_1-x_3\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_2-x_4\,.\,x_3-x_4)\,(c-d\,.\,d-b\,.\,b-c)\,A_m$$

$$=\ A_m\,\Sigma\,[c-d\,.\,B_{12}^x\,BE_a^1C_a\,D_a]\ +\ A_{12}\,\Sigma\,[c-d\,.\,B_{12}^x\,BE_a^2C_a\,D_a],$$

where, as before, the summations refer each to the three terms obtained by the cyclical interchange of the letters b, c, d; these being any three of the five letters other than a: and the remaining two letters e, f enter into the formulae symmetrically. The formula thus gives for A_m ten values which are of course equal to each other.

By reason of the undetermined factor $\dfrac{\sqrt{\mu}}{\sqrt{\lambda}}$, the formula gives only the proportional value of A_m; viz. combining it with the like formulae for B_m, &c., we have determinate values of the ratios $A_m:B_m:\ldots:F_{56}$. But this being understood, we regard the formula as a formula for each single-letter function of x_3, x_4 in terms of the single and double-letter functions of x_1, x_2 and of x_3, x_4 respectively.

163. We require further the expressions for the double-letter functions of x_3, x_4.

Consider for example the function DE_m, which is

$$= \frac{1}{x_3-x_4}\big[\sqrt{d_3\,e_3\,f_3\,a_4\,b_4\,c_4}-\sqrt{d_4\,e_4\,f_4\,a_3\,b_3\,c_3}\big];$$

then multiplying by $A_m\,B_m\,C_m=\sqrt{a_3\,b_3\,c_3\,a_4\,b_4\,c_4}$, we have

$$DE_m\,A_m\,B_m\,C_m\ =\ \frac{1}{x_3-x_4}\big[a_3b_3c_3\sqrt{X_4}-a_4b_4c_4\sqrt{X_3}\big],$$

or recollecting that $\sqrt{\lambda}\,\sqrt{X_3}$ and $\sqrt{\lambda}\,\sqrt{X_4}$ are $=\alpha x_3^3+\beta x_3^2+\gamma x_3+\delta$ and $\alpha x_4^3+\beta x_4^2+\gamma x_4+\delta$ respectively, this may be written

$$\sqrt{\lambda}\,DE_m\,A_mB_mC_m\ =\ \frac{1}{x_3-x_4}\big\{(a-x_3\,.\,b-x_3\,.\,c-x_3)(\alpha x_4^3+\beta x_4^2+\gamma x_4+\delta)$$

$$-(a-x_4\,.\,b-x_4\,.\,c-x_4)(\alpha x_3^3+\beta x_3^2+\gamma x_3+\delta)\big\}.$$

Using the well-known identity

$$\alpha x^3+\beta x^2+\gamma x+\delta\ =\ \Sigma\,\frac{\alpha a^3+\beta a^2+\gamma a+\delta}{a-b\,.\,a-c\,.\,a-d}\,(b-x\,.\,c-x\,.\,d-x),$$

where the summation extends to the four terms obtained by the cyclical interchanges of the letters a, b, c, d: and the like identity for $\alpha x^3+\beta x^2+\gamma x+\delta$, there will be terms in $\alpha a^3+\beta a^2+\gamma a+\delta$, $\alpha b^3+\beta b^2+\gamma b+\delta$, $\alpha c^3+\beta c^2+\gamma c+\delta$, but the term in $\alpha d^3+\beta d^2+\gamma d+\delta$ will disappear of itself. After some easy reductions, the result is

$$\sqrt{\lambda}\, DE_m\, A_m B_m C_m \;=\; \Sigma\, \frac{\alpha a^3+\beta a^2+\gamma a+\delta}{b-a\,.\,c-a\,.\,d-a}\, B^x_{12} C^x_{12},$$

where the summation extends to the three terms obtained by the cyclical interchanges of the letters a, b, c: We have $\alpha a^3+\beta a^2+\gamma a+\delta=\sqrt{\mu}\, A_a A_m A_m$ and similarly for the other two terms; the whole equation thus divides by $A_m B_m C_m$, and we find

$$-\frac{\sqrt{\mu}}{\sqrt{\lambda}}\,(x_1-x_3\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_2-x_4)\, DE_m \;=\; \frac{1}{x_1-x_2\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_3-x_4}\, M,$$

where M is a given rational and integral function of the single and double-letter functions of x_1, x_2 and x_3, x_4. The factor on the left-hand side has been made the same as in the formula for the single-letter functions A_m, &c., and to do this it was necessary to bring in on the right-hand side the factor

$$\frac{1}{x_1-x_2\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_3-x_4};$$

this disappears in the expression for the ratio of two double-letter functions; but it enters into the expression for the ratio of a single-letter to a double-letter function, and it then requires to be itself expressed in terms of the functions of x_1, x_2 and x_3, x_4: it is easy to see that we have

$$\frac{1}{x_1-x_2\,.\,x_1-x_4\,.\,x_2-x_3\,.\,x_3-x_4} \;=\; \Sigma\, \frac{(A^x_{12}B^x_{12}-A^x_{34}B^x_{34})\,(A^x_{12}C^x_{12}-A^x_{34}C^x_{34})}{(a-b)(a-c)},$$

where the summation extends to the three terms obtained by the cyclical interchanges of the letters a, b, c: these being a set of any three out of the six letters.

We have, in what precedes, obtained the expressions for the ratios of the 16 functions $A_m, \ldots, F_{56}, AB_m, \ldots, DE_m$ in terms of the ratios of the like functions of x_1, x_2 and x_3, x_4.

CHAPTER VII. THE FUNCTIONS T, U, V, Θ.

The present chapter is substantially a reproduction of C. and G.'s seventh section, "Die Function T^{κ}_{μ}," borrowing only from the next section the definition of the theta-function; but for greater simplicity I consider for the most part, the case, fixed curve a quartic, $n = 4$, $p = 3$.

Integral Form of the Affected Theorem. Art. Nos. 164 to 169.

164. Writing for shortness $\dfrac{(x, y, z)_0^{n-2}\, d\omega}{012} = d\Pi_{12}$, we are concerned with the integrals $\displaystyle\int_{a'}^{a} d\Pi_{12}$ which present themselves in connexion with the affected theorem: the notation is explained, Chap. V.; a, a' are points on the curve f; the variable may be any parameter serving for the determination of the current point, and the integral, taken from the value which belongs to the point a' to the value which belongs to the point a, is represented as above by means of the two points a, a' as limits of the integral. It is assumed that the integral is a canonical integral having the limits and the parametric points interchangeable, $\displaystyle\int_{a'}^{a} d\Pi_{12} = \int_{2}^{1} d\Pi_{aa'}$: see Chapter IV.

165. Writing for shortness

$$\left(\int_{a'}^{a} + \int_{b'}^{b} + \int_{c'}^{c} + \ldots\right) d\Pi_{12} \;=\; \int \begin{pmatrix} a,\, b,\, c, \ldots \\ a',\, b',\, c', \ldots \end{pmatrix} d\Pi_{12},$$

then if ϕ, ψ are curves each of the order m, the former of them intersecting the fixed curve f in the points a, b, c, ..., and the latter of them intersecting the same curve in the points a', b', c',..., and if ϕ_1, ϕ_2, ψ_1, ψ_2 are what the functions ϕ, ψ become on substituting therein in place of the current coordinates the values which belong to the parametric points 1, 2 respectively; the theorem becomes

$$\int \begin{pmatrix} a,\, b,\, c, \ldots \\ a',\, b',\, c', \ldots \end{pmatrix} d\Pi_{12} \;=\; \log \frac{\phi_1 \psi_2}{\phi_2 \psi_1}.$$

The superior limits may be interchanged in any manner, and so also the inferior limits may be interchanged in any manner. If a superior limit coincide with an inferior limit, the two may be thus considered as belonging to an integral which will then have the value 0, and the coincident points may therefore be omitted from the expression on the left-hand side: and so in the case of any number of coincidences.

166. If the intersections of the curves ϕ, ψ and the parametric points are situate on a curve of the order m; then taking the equation of this curve to be $\phi + \lambda\psi = 0$, we have simultaneously $\phi_1 + \lambda\psi_1 = 0$, $\phi_2 + \lambda\psi_2 = 0$; whence $\phi_1\psi_2 = \phi_2\psi_1$, and the logarithmic term disappears: viz. the theorem becomes

$$\int \begin{pmatrix} a,\, b,\, c, \ldots \\ a',\, b',\, c', \ldots \end{pmatrix} d\Pi_{12} \;=\; 0.$$

167. Suppose that the curves ϕ, ψ are each of them a major curve, that is, a curve of the order $n-2$ passing through the δ dps, and consequently besides meeting the curve f in $n(n-2)-2\delta$, $=2p+n-2$ points: the theorem is

$$\int \begin{pmatrix} a,\ b,\ c,\ldots \\ a',\ b',\ c',\ldots \end{pmatrix} d\Pi_{12} \;=\; \log \frac{\phi_1\psi_2}{\phi_2\psi_1},$$

where the numbers of the superior and of the inferior points are each $=2p+n-2$.

168. Suppose further that the curves ϕ, ψ, being major curves as above, pass each of them through the $n-2$ residues of 1, 2; they besides meet in $(n-2)(n-3)$ points, viz. these are the δ dps and $(n-2)(n-3)-\delta$ variable points: these $(n-2)(n-3)$ points lie on a minor curve, that is, a curve of the order $n-3$ passing through the dps; and the minor curve together with the parametric line 12 make together a major curve, passing through the intersections of ϕ, ψ and also through the parametric points 1, 2: viz. these points and the intersections of ϕ, ψ are situate on a curve of the order $n-2$; the logarithmic term thus vanishes. The intersections of ϕ with the fixed curve are the δ dps, the $n-2$ residues and $2p$ other points, say these are a, b, c,..., a^x, b^x, c^x,...; similarly the curve ψ meets the fixed curve in the δ dps, the $n-2$ residues, and in $2p$ other points, say these are d, e, f,..., d^x, e^x, f^x,...: the theorem is

$$\int \begin{pmatrix} a,\ b,\ c,\ldots;\ a^x,\ b^x,\ c^x,\ldots \\ d,\ e,\ f,\ldots;\ d^x,\ e^x,\ f^x,\ldots \end{pmatrix} d\Pi_{12} \;=\; 0,$$

where there are $2p$ superior and inferior points respectively.

169. I introduce the definitions: a minor curve meets the fixed curve in the dps and in $2p-2$ other points, called "cominors": a major curve passing through the $n-2$ residues of the points 1, 2, meets the fixed curve in the δ dps, the $n-2$ residues and in $2p$ other points, called "comajors in regard to the points 1, 2." Observe that $p-1$ of the cominors determine uniquely the remaining $p-1$ cominors; and similarly p of the comajors determine uniquely the remaining p comajors.

The foregoing theorem thus is that the sum $\int (\ )\, d\Pi_{12}=0$, when the superior points and the inferior points are each of them a system of comajors in regard to the parametric points 1, 2.

Fixed Curve a Quartic. Art. No. 170.

170. It would be easy to go on with the general form; but as already mentioned, I prefer to consider the case, fixed curve a quartic, $n=4$, $p=3$. A minor curve is here a line meeting the quartic in 4 points, which are "cominors"; the major curve is a conic, and if this passes through the residues of 1, 2 it besides meets the quartic in 6 points, which are "comajors in regard to the points 1, 2." Two points and their residues are cominors, but this is only by reason that $n-3=1$.

The Function T. Art. No. 171.

171. In conformity with C. and G., I introduce the functional symbol

$$T_{12}\begin{pmatrix} a,\ b,\ c,\ldots \\ a',\ b',\ c',\ldots \end{pmatrix} \;=\; \int \begin{pmatrix} a,\ b,\ c,\ldots \\ a',\ b',\ c',\ldots \end{pmatrix} d\Pi_{12},$$

so that T denotes a function of the parametric points, and of the sets of superior and inferior points respectively. The foregoing theorem for the quartic thus is

$$T_{12}\begin{pmatrix} a,\ b,\ c,\ a^x,\ b^x,\ c^x \\ d,\ e,\ f,\ d^x,\ e^x,\ f^x \end{pmatrix} = 0.$$

Observing that

$$\int_d^a + \int_{d^x}^{a^x} = 2\int_2^1 - \int_{a^x}^d + \int_{d^x}^a,$$

and so in other cases, this may be written

$$T_{12}\begin{pmatrix} a,\ b,\ c \\ d,\ e,\ f \end{pmatrix} = T_{12}\begin{pmatrix} a^x,\ b^x,\ c^x \\ a,\ b,\ c \end{pmatrix} - T_{12}\begin{pmatrix} d,\ e,\ f \\ d^x,\ e^x,\ f^x \end{pmatrix};$$

and if, as a definition of $T_{12}(a, b, c)$, we write

$$T_{12}(a, b, c) = T_{12}\begin{pmatrix} a,\ b,\ c \\ a^x,\ b^x,\ c^x \end{pmatrix} - 2\int_2^1,$$

where a^x, b^x, c^x are the comajors of a, b, c in regard to 1, 2, then the equation is

$$T_{12}\begin{pmatrix} a,\ b,\ c \\ d,\ e,\ f \end{pmatrix} = T_{12}(a, b, c) - T_{12}(d, e, f),$$

viz. the function of the $(2p + 2 =)8$ points 1, 2, a, b, c, d, e, f is here expressed as a difference of two functions each of $(p+2 =)5$ points: $T_{12}(a, b, c)$ is regarded as a function of the 5 points 1, 2, a, b, c, because the remaining points a^x, b^x, c^x depend only on these 5 points.

The Function U. Art. Nos. 172 to 175.

172. We consider on the quartic the points ξ, μ; 1, 2, 3; and take f, f' for the cominors of 2, 3; g, g' for the cominors of 3, 1; and h, h' for the cominors of 1, 2. We write

$$T = T_\mu(1, 2, 3),$$

it is to be shown that there exists a function $U(1, 2, 3; \xi)$ such that

$$\delta U = \tfrac{1}{2}\left[\delta_\xi T + \delta_1 T_1 + \delta_2 T_2 + \delta_3 T_3\right],$$

viz. considering ξ, 1, 2, 3 as variable points on the quartic, the whole infinitesimal variation of U is the sum of these parts, where $\delta_\xi T$ is the variation of T when only ξ is varied, $\delta_1 T_1$ the variation of T_1 when only 1 is varied, and similarly for $\delta_2 T_2$ and $\delta_3 T_3$. We consider in the proof three other points 4, 5, 6 on the quartic; and taking l, l' for the cominors of 5, 6; m, m' for those of 6, 4; and n, n' for those of 4, 5, we write further

$$X_1 = T_\mu(4, l, l'), \qquad X_2 = T_\mu(5, m, m'), \qquad X_3 = T_\mu(6, n, n').$$

and it then requires to be shown that

$$\begin{aligned}
\tfrac{1}{2}\,\delta T &= \int \binom{123}{456} d\Pi_{1\mu} + \tfrac{1}{2}\,T_{1\mu}(4,5,6),\\
\tfrac{1}{2}(T_1 - X_1) &= \int \binom{456}{423} d\Pi_{1\mu} + \log \frac{\mu 23\,.\,156}{\mu 56\,.\,123},\\
\tfrac{1}{2}(T_2 - X_2) &= \int \binom{564}{531} d\Pi_{1\mu} + \log \frac{\mu 31\,.\,264}{\mu 64\,.\,231},\\
\tfrac{1}{2}(T_3 - X_3) &= \int \binom{645}{612} d\Pi_{1\mu} + \log \frac{\mu 12\,.\,345}{\mu 45\,.\,312},
\end{aligned}$$

where $\mu 23$ is the determinant formed with the coordinates of the points μ, 2, 3 respectively; and so in other cases.

173. We have

$$T_{1\mu}\binom{1,\,2,\,3}{4,\,5,\,6} = \tfrac{1}{2}\,T_{1\mu}(1,2,3) - \tfrac{1}{2}\,T_{1\mu}(4,5,6),$$

that is,

$$= \tfrac{1}{2}\,T - \tfrac{1}{2}\,T_{1\mu}(4,5,6),$$

and thence the above value of $\frac{1}{2}\,\delta T$.

The affected theorem gives

$$\int \binom{f,\,f',\,2,\,3}{l,\,l',\,5,\,6} d\Pi_{1\mu} = \log \frac{F_1 L_\mu}{F_\mu L_1},$$

where $F = 0$ is the equation of the line through f, f', 2, 3; and F_1, F_μ are what the function F becomes, on substituting therein for the current coordinates the coordinates of the points 1, μ respectively. And similarly $L = 0$ is the equation of the line through l, l', 5, 6; and L_1, L_μ are what the function L becomes by the same sub stitutions respectively. The values of F_1, F_μ are 123, $\mu 23$: those of L_1, L_μ are 156, $\mu 56$: and the logarithmic term is thus

$$= \log \frac{\mu 23\,.\,156}{\mu 56\,.\,123}.$$

We then have

$$\tfrac{1}{2}(T_1 - X_1) = \int \binom{f,\,f',\,2,\,3}{4,\,l,\,l'} d\Pi_{1\mu} = \int \binom{f,\,f',\,2,\,3}{l,\,l',\,5,\,6} d\Pi_{1\mu},$$

and in this last expression for $\frac{1}{2}(T_1 - X_1)$ substituting for the second term the logarithmic value just obtained, we have the required expression for $\frac{1}{2}(T_1 - X_1)$: and those for $\frac{1}{2}(T_2 - X_2)$ and $\frac{1}{2}(T_3 - X_3)$ are deduced by mere cyclical permutations of the letters.

174. Returning to the assumed relation $\delta U = \frac{1}{2}\,[\delta_\xi T + \delta_1 T_1 + \delta_2 T_2 + \delta_3 T_3]$, in order to the existence of the function U, it is only necessary to show that $T - T_1$ contains no term in 1, ξ, that is, no term depending on both these points, and that $T_1 - T_2$ contains no term in 1, 2; for then, by symmetry, the like properties hold in regard to $T - T_2$, $T - T_3$, $T_1 - T_3$, $T_2 - T_3$ respectively, and the assumed expression is a complete differential, from which the function U may be obtained by integration.

175. To show that $T - T_1$ contains no term in 1, ξ.

For T, the only term in 1, ξ is $\int_1^\xi d\Pi_{1\mu}$,

" T_1 " " " $\int_1^\xi d\Pi_{1\mu}$,

and it is to be shown that the difference of the two integrals contains no term in 1, ξ. Considering on the quartic the two new points δ, e, the first integral is

$$\int \begin{pmatrix} 1, \xi \\ \delta, e \end{pmatrix} (d\Pi_{1\mu} + d\Pi_{1\mu}) = \int_\delta^1 d\Pi_{1\mu} + \int_e^\xi d\Pi_{1\mu} + \int_1^\xi d\Pi_{1\mu},$$

and the second is

$$\int \begin{pmatrix} \xi, e \\ \delta, 4 \end{pmatrix} (d\Pi_{1\mu} + d\Pi_{1\mu}) = \int_\delta^\xi d\Pi_{1\mu} + \int_4^e d\Pi_{1\mu} + \int_1^4 d\Pi_{1\mu}.$$

Hence in the difference the only terms which can contain 1, ξ are

$$\int_e^1 d\Pi_{1\mu} + \int_\xi^4 d\Pi_{1\mu},$$

and this is $= 0$: wherefore there is not in the difference any term in 1, ξ. This proves the property for $T - T_1$. The property for $T_1 - T_2$ is proved in a similar manner.

Theorems in regard to the Function U. Art. Nos. 176 to 179.

176. Theorem (A). To prove

$$U(1,2,3;\xi) - U(1,2,3;\mu) = \tfrac{1}{2} T_{\xi\mu}(1,2,3), \tag{A}$$

we have

$$\begin{aligned} U(1,2,3;\xi) - U(1,2,3;\mu) &= \int_\mu^\xi d_\xi U = \tfrac{1}{2}\int_\mu^\xi d_\xi T \\ &= \tfrac{1}{2} T_\mu(1,2,3) - \tfrac{1}{2} T_\mu(1,2,3), \end{aligned}$$

and $T_{\xi\mu}(1,2,3) = \int \begin{pmatrix} 1, 2, 3 \\ 1^x, 2^x, 3^x \end{pmatrix} d\Pi_{\xi\mu}$, where $d\Pi_{\xi\mu} = 0$, viz. considering this as derived from $d\Pi_{\xi\mu}$, $= Q_{\xi\mu} d\omega$, by making the point ξ coincide with μ, then when ξ is indefinitely near to μ, the numerator and denominator of $Q_{\xi\mu}$ are each of them infinitesimal of the orders 3 and 2 respectively, and thus the function $Q_{\xi\mu}$ ultimately vanishes (see as to this, Chap. V. Art. Nos. 99 to 106). We have therefore $T_{\xi\mu}(1,2,3) = 0$, and the required theorem is proved.

177. Theorem (B). To prove

$$U(1,2,3;\xi) - U(4,2,3;\xi) = \tfrac{1}{2} T_{1\xi}(f, f'), \tag{B}$$

where, as before, f, f' are the cominors of 2, 3, that is, 2, 3, f, f' lie on a line, we have

$$\begin{aligned} U(1,2,3;\xi) - U(4,2,3;\xi) &= \int_4^1 d_1 U = \tfrac{1}{2}\int_4^1 d_1 T \\ &= \tfrac{1}{2} T_\mu(\xi, f, f') - \tfrac{1}{2} T_{4\mu}(\xi, f, f'), \end{aligned}$$

the point μ is arbitrary, and it may be taken to coincide with 4; but we then have $T_{4\mu}(\xi, f, f') = 0$, and the theorem is thus proved.

178. Theorem (C). We have

$$T_{1\xi^x}(\xi, f, f') + T_{1\xi^x}(\eta, k, k') \;=\; 0; \tag{C}$$

where ξ, η, 1, 2, 3 are arbitrary points on the quartic; 1^x, 2^x, 3^x are the comajors of 1, 2, 3 in regard to ξ, η, viz. the points 1, 2, 3, 1^x, 2^x, 3^x lie on a conic which passes through ξ, η: the residues (or cominors) of ξ, η: f, f' are the cominors of 2, 3; and k, k' are the cominors of 2^x, 3^x.

Taking θ, θ' for the cominors of 1, 1^x, the four lines $\xi\eta\theta\theta'$, $11^x\theta\theta'$, $23ff'$ and $2^x3^xkk'$ form a quartic cutting the fixed quartic in the 16 points: but of these, ξ, η, 1, 2, 3, 1^x, 2^x, 3^x lie in a conic: hence the remaining 8 points θ, θ', ξ, η, f, f', k, k' lie in a conic: that is, ξ, η, f, f', k, k' lie on a conic through θ, θ', the residues of 1, 1^x: or they are comajors in regard to 1, 1^x; whence the theorem.

179. We have

From A. *From B.*

$$\begin{aligned}\tfrac{1}{2}T_{1\xi^x}(\xi, f, f') &= U(f, f'; 1) - U(f, f'; 1^x) = U(1, 2, 3; \xi) - U(1^x, 2, 3; \xi),\\ \tfrac{1}{2}T_{1\xi^x}(\eta, k, k') &= U(\eta, k, k'; 1) - U(\eta, k, k'; 1^x) = U(1, 2^x, 3^x; \eta) - U(1^x, 2, 3; \eta);\end{aligned}$$

viz. we have thus two expressions for each term of the equation (C),

$$T_{1\xi^x}(\xi, f, f') + T_{1\xi^x}(\eta, k, k') \;=\; 0.$$

In particular, we have Theorem (D)

$$U(1, 2, 3; \xi) - U(1^x, 2, 3; \xi) \;=\; -U(1, 2^x, 3^x; \eta) + U(1^x, 2^x, 3^x; \eta). \tag{D}$$

Again, we have $T_{1\xi}(1, 2, 3) + T_{1\xi}(1^x, 2^x, 3^x) = 0$; where 1, 2, 3, 1^x, 2^x, 3^x are comajors in regard to ξ, η: and

$$\begin{aligned}\tfrac{1}{2}\,T_{\xi\eta}(1, 2, 3) &= U(1, 2, 3; \xi) - U(1, 2, 3; \eta),\\ \tfrac{1}{2}\,T_{\xi\eta}(1^x, 2^x, 3^x) &= U(1^x, 2^x, 3^x; \xi) - U(1^x, 2^x, 3^x; \eta);\end{aligned}$$

whence Theorem (E),

$$U(1, 2, 3; \xi) - U(1, 2, 3; \eta) \;=\; -U(1^x, 2^x, 3^x; \xi) + U(1^x, 2^x, 3^x; \eta). \tag{E}$$

The Function V. Art. Nos. 180 to 182.

180. It is convenient to consider U as a logarithm, say

$$-U(1, 2, 3; \xi) \;=\; \log V(1, 2, 3; \xi), \qquad \text{or} \qquad V(1, 2, 3; \xi) = \exp . {-U}(1, 2, 3; \xi);$$

V, like U, is a function of the $(p + 1 =)\,4$ points 1, 2, 3, ξ, on the quartic.

The equation (D) thus becomes

$$\frac{V(1, 2, 3; \xi)\, V(1^x, 2, 3; \xi)}{V(1^x, 2^x, 3^x; \eta)\, V(1, 2^x, 3^x; \eta)},$$

where 1, 2, 3, 1^x, 2^x, 3^x are comajors in regard to ξ, η: the equation shows that, in the function $\dfrac{V(1,2,3;\xi)}{V(1^x,2^x,3^x;\eta)}$, we can without alteration of the value interchange a pair of points 1, 1^x out of the system of comajor points; and it of course follows that we can in any manner whatever interchange these points, so as to have any three of them in the numerator-function and the remaining three in the denominator-function. In particular, we have

$$\frac{V(1,2,3;\xi)}{V(1^x,2^x,3^x;\eta)} \;=\; \frac{V(1^x,2^x,3^x;\xi)}{V(1,2,3;\eta)}.$$

The equation (E) becomes

$$\frac{V(1,2,3;\xi)\,V(1^x,2^x,3^x;\eta)}{V(1,2,3;\eta)\,V(1^x,2^x,3^x;\xi)},$$

and multiplying we find

$$V(1,2,3;\xi)^2 \;=\; V(1^x,2^x,3^x;\eta)^2,$$

that is, $V(1,2,3;\xi) = \pm V(1^x,2^x,3^x;\eta)$, the sign being determinately + or determinately −, according to the precise definition of the function V.

181. Considering ξ as fixed points on the curve but η as also; and 1^x, 2^x, 3^x a variable point (that is, the parametric line $\xi\eta$ as rotating about the fixed point η), the points 1, 2, 3 are then determined as the remaining intersections with the quartic of the conic which passes through the points 1^x, 2^x, 3^x and through the points ξ, η, which are the residues of ξ, η. And by the theorem just obtained it appears that 1, 2, 3, thus varying, the function $V(1,2,3;\xi)$ remains constant. This comes to saying that V, considered as a function of the points 1, 2, 3, ξ, satisfies a certain linear partial differential equation of the first order, having a solution $V = F(u,v,w)$, an arbitrary function of u, v, w, determinate functions of the points 1, 2, 3, ξ. And if we can find u, v, w functions of these points such that they each of them remain constant when the points 1, 2, 3, ξ vary as above, then the arbitrary function of u, v, w will remain constant for the variation in question and will thus be a value of the function V.

182. It is easily seen that such functions are

$$u,\ v,\ w \;=\; \left(\int^1 + \int^2 + \int^3 - \int^\xi\right) x\,d\omega,\quad y\,d\omega,\quad z\,d\omega,$$

the inferior limits being given points which are regarded as absolute constants. For by the pure theorem, we have

$$\Sigma\,(x,y,z)_1\,d\omega \;=\; 0,$$

where $(x,y,z)_1$ is an arbitrary linear function, and where the summation extends to all the intersections of the quartic with any given curve. Writing

$$p \;=\; \int x\,d\omega,\quad \int y\,d\omega \ \text{ or } \int z\,d\omega,\quad \text{that is,}\quad p = \int^1 x\,d\omega,\ \int^1 y\,d\omega \text{ or } \int^1 z\,d\omega,$$

the inferior limits being any absolutely fixed point on the curve, and similarly p_2, &c.; the integral form of the theorem is $\Sigma p =$ constant. And applying the theorem successively to the parametric line, and to the conic which determines the points 1, 2, 3, we have

$$p_\xi + p_\eta + p_{\xi'} + p_{\eta'} = \text{const.},$$
$$p_\xi + p_\eta + p_1 + p_2 + p_3 + p_{1^x} + p_{2^x} + p_{3^x} = \text{const.}$$

Taking the difference of these equations, we have

$$p_1 + p_2 + p_3 - p_{\xi'} + p_{1^x} + p_{2^x} + p_{3^x} - p_{\eta'} \;=\; \text{const.},$$

viz. the points η, 1^x, 2^x, 3^x being fixed points, this is

$$p_1 + p_2 + p_3 - p_\xi \;=\; \text{const.},$$

that is, the functions u, v, w defined as above are each of them constant under the variation in question.

The Function Θ. Art. Nos. 183 and 184.

183. The function $V(1,2,3;\xi)$ of the $(p+1=)\,4$ points 1, 2, 3, ξ, is thus a function of the $(p=)\,3$ arguments

$$u,\ v,\ w \;=\; \left(\int^1 + \int^2 + \int^3 - \int^\xi\right) x\,d\omega,\quad y\,d\omega,\quad z\,d\omega.$$

Disregarding a constant and exponential factor, we say that it is a theta-function of these arguments, and we write the result provisionally in the form

$$V(1,2,3;\xi) \;=\; \Theta(u,v,w),$$

the more precise definition of the theta-function being reserved for further consideration.

184. It appears by what precedes that a sum of $(p=)\,3$ integrals $\int \begin{pmatrix} abc \\ def \end{pmatrix} d\Pi_{12}$, otherwise called $T_{12}\binom{a,\,b,\,c}{d,\,e,\,f}$, is in the first place expressed (see No. 171) as a difference, $= \frac{1}{2}\,T_{12}(a,b,c) - \frac{1}{2}\,T_{12}(d,e,f)$, of two functions T. Each of these is by Theorem (A) (No. 176) expressed as a difference of two functions U, that is, as the difference of the logarithms, or logarithm of the quotient, of two functions V: such function V according to its original definition a function of $(p+1=)\,4$ points, but in such wise that the function is expressible as a function of $(p=)\,3$ arguments, and so expressed it is a Θ-function of these arguments: and the final result thus is that the sum of $(p=)\,3$ integrals $\int \begin{pmatrix} a,b,c \\ d,e,f \end{pmatrix} d\Pi_{12}$ is equal to the logarithm of a fraction, whereof the numerator and the denominator are each of them a product of two Θ-functions.

826.

Note on a Partition-Series.

[From the *American Journal of Mathematics*, vol. vi. (1884), pp. 63, 64.]

Prof. Sylvester, in his paper, "A Constructive theory of Partitions, &c.," *American Journal of Mathematics*, vol. v. (1883), p. 282, has given the following very beautiful formula

$$(1+ax)(1+ax^2)(1+ax^3)\ldots \;=\; 1+\frac{(1+ax^2)\,xa}{1-x}+\frac{1}{1-x\,.\,1-x^2}(1+ax)(1+ax^3)\,x^4a^2$$

$$+\frac{1}{1-x\,.\,1-x^2\,.\,1-x^3}(1+ax)(1+ax^2)(1+ax^4)\,x^{12}a^3+\ldots,$$

or, as this may be written,

$$\Omega \;=\; 1+P+Q(1+ax)+R(1+ax)(1+ax^2)+S(1+ax)(1+ax^2)(1+ax^3)+\ldots$$

where

$$P=\frac{(1+ax^2)\,xa}{\mathbf{1}},\qquad Q=\frac{(1+ax^4)\,x^5a^2}{\mathbf{1.2}},\qquad R=\frac{(1+ax^6)\,x^{12}a^3}{\mathbf{1.2.3}},\qquad S=\frac{(1+ax^8)\,x^{22}a^4}{\mathbf{1.2.3.4}},\ \text{\&c.},$$

the heavy figures **1**, **2**, **3**, **4**, ... of the denominators being, for shortness, written to denote $1-x$, $1-x^2$, $1-x^3$, $1-x^4$, ... respectively. The x-exponents 1, 5, 12, 22, ... are the pentagonal numbers $\frac{1}{2}(3n^2-n)$.

To prove this, writing

$$P'=\frac{ax^2}{\mathbf{1}},\quad Q'=\frac{ax^4}{\mathbf{1}}+\frac{a^2x^6}{\mathbf{1.2}},\quad R'=\frac{ax^6}{\mathbf{1}}+\frac{a^2x^9}{\mathbf{1.2}}+\frac{a^3x^{12}}{\mathbf{1.2.3}},\quad S'=\frac{ax^8}{\mathbf{1}}+\frac{a^2x^{11}}{\mathbf{1.2}}+\frac{a^3x^{15}}{\mathbf{1.2.3}}+\frac{a^4x^{18}}{\mathbf{1.2.3.4}},\ \text{\&c.},$$

where the x-exponents are

$$2;\quad 3,\ 3+4;\quad 4,\ 4+5,\ 4+5+6;\quad 5,\ 5+6,\ 5+6+7,\ 5+6+7+8;\ \text{\&c.},$$